

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 07

DATE : 01-12-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 07

Topics

DATE : 01 - 12 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : Center of Mass, Collision

CHEMISTRY : Equilibrium

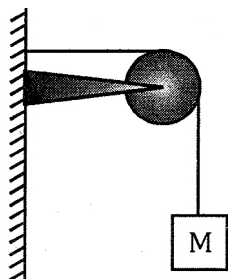
ZOOLOGY : Digestive System, Immune System

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

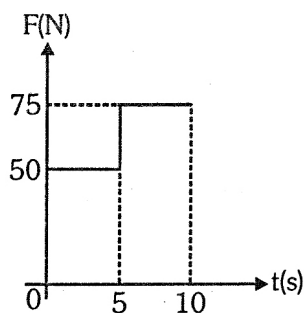
BEWARE OF NEGATIVE MARKING

1. A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by :-



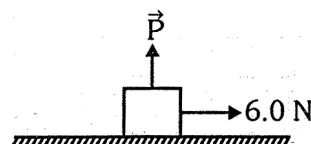
- (1) $\sqrt{2} Mg$
(2) $\sqrt{2} mg$
(3) $\left(\sqrt{(M+m)^2 + m^2}\right)g$
(4) $\left(\sqrt{(M+m)^2 + M^2}\right)g$

2. A force represented as show in figure acts on a body having a mass of 16 kg. The velocity of the body at $t = 10$ s, if the body starts from rest:-



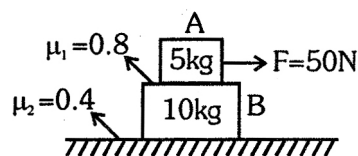
- (1) 100 m/s (2) 50 m/s
(3) 49 m/s (4) 39 m/s

3. A 2.5 kg block is initially at rest on a horizontal surface. A 6.0 N horizontal force and a vertical force $\vec{P}$ are applied to the block as shown in figure. The coefficient of friction for the block and surface is 0.4. The magnitude of friction force when $P = 9\text{ N}$ ($g = 10 \text{ m/s}^2$)



- (1) 6.0 N (2) 6.4 N
(3) 9.0 N (4) zero

4. Find acceleration of block A and B. If $\mu_1 = 0.8$ and $\mu_2 = 0.4$



- (1) 0 m/s^2 and 0 m/s^2
(2) 1 m/s^2 and $1/2 \text{ m/s}^2$
(3) 2 m/s^2 and 0 m/s^2
(4) $1/2 \text{ m/s}^2$ and $1/4 \text{ m/s}^2$

5. If K.E. increases by 4%. Then momentum will increase by :-

- (1) 1.5% (2) 9% (3) 3% (4) 2%

6. A force $\vec{F} = (3x^2 + 2x - 7) \text{ N}$ acts on a 2 kg body as a result of which the body gets displaced from $x = 0$ to $x = 5 \text{ m}$. The work done by the force will be-

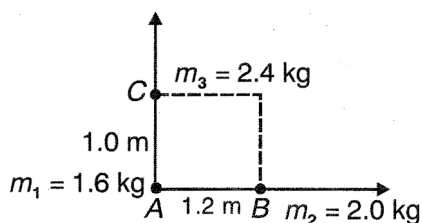
- (1) 35 J (2) 70 J
(3) 115 J (4) 270 J

SPACE FOR ROUGH WORK

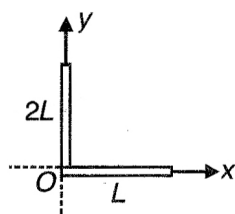
7. A string breaks if its tension exceeds 10 newtons. A stone of mass 250 gm tied to this string of length 10 cm is rotated in a horizontal circle. The maximum angular velocity of rotation can be :-
 (1) 20 rad/s (2) 40 rad/s
 (3) 100 rad/s (4) 200 rad/s
8. What is the value of linear velocity, if $\vec{\omega} = 3\hat{i} - 4\hat{j} + \hat{k}$ and $\vec{r} = 5\hat{i} - 6\hat{j} + 6\hat{k}$
 (1) $6\hat{i} + 2\hat{j} - 3\hat{k}$
 (2) $-18\hat{i} - 13\hat{j} + 2\hat{k}$
 (3) $4\hat{i} - 13\hat{j} + 6\hat{k}$
 (4) $6\hat{i} - 2\hat{j} + 8\hat{k}$
9. A machine which is 75 percent efficient, uses 12 joules of energy in lifting up a 1 kg mass through a certain distance. The mass is then allowed to fall through that distance. The velocity at the end of its fall is (in ms^{-1})
 (1) $\sqrt{24}$ (2) $\sqrt{32}$
 (3) $\sqrt{18}$ (4) $\sqrt{9}$
10. A force of $2\hat{i} + 3\hat{j} + 4\hat{k}$ N acts on a body for 4 second, produces a displacement of $(3\hat{i} + 4\hat{j} + 5\hat{k})$ m. The power used is
 (1) 9.5 W (2) 7.5 W
 (3) 6.5 W (4) 4.5 W
11. Two equal masses m_1 and m_2 moving along the same straight line with velocities + 3 m/s and - 5 m/s respectively collide elastically. Their velocities after the collision will be respectively
 (1) + 4 m/s for both
 (2) - 3 m/s and +5 m/s
 (3) - 4 m/s and + 4 m/s
 (4) - 5 m/s and + 3 m/s
12. A body falling from a height of 10m rebounds from hard floor. If it loses 20% energy in the impact, then coefficient of restitution is
 (1) 0.89 (2) 0.56
 (3) 0.23 (4) 0.18
13. A tennis ball dropped from a height of 2 m rebounds only 1.5 m after hitting the ground. What fraction of its energy is lost in the impact
 (1) $\frac{1}{4}$ (2) $\frac{1}{2}$
 (3) $\frac{1}{3}$ (4) $\frac{1}{8}$
14. A body of mass m moving with velocity v makes a head-on collision with another body of mass $2m$ which is initially at rest. The loss of kinetic energy of the colliding body (mass m) is
 (1) $\frac{1}{2}$ of its initial kinetic energy
 (2) $\frac{1}{9}$ of its initial kinetic energy
 (3) $\frac{8}{9}$ of its initial kinetic energy
 (4) $\frac{1}{4}$ of its initial kinetic energy

SPACE FOR ROUGH WORK

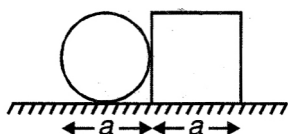
15. Three point masses m_1 , m_2 and m_3 are placed at the corners of a thin massless rectangular sheet ($1.2 \text{ m} \times 1.0 \text{ m}$) as shown. Centre of mass will be located at the point



- (1) $(0.8, 0.6) \text{ m}$ (2) $(0.6, 0.8) \text{ m}$
(3) $(0.4, 0.4) \text{ m}$ (4) $(0.5, 0.6) \text{ m}$
16. Figure shows a composite system of two uniform rods of lengths as indicated. Then the coordinates of the centre of mass of the system of rods are

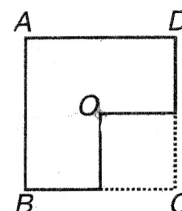


- (1) $\left(\frac{L}{2}, \frac{2L}{3}\right)$ (2) $\left(\frac{L}{4}, \frac{2L}{3}\right)$
(3) $\left(\frac{L}{6}, \frac{2L}{3}\right)$ (4) $\left(\frac{L}{6}, \frac{L}{3}\right)$
17. A circular plate of diameter 'a' is kept in contact with a square plate of side a as shown. The density of the material and the thickness are same everywhere. The centre of mass of composite system will be



- (1) Inside the circular plate
(2) Inside the square plate
(3) At the point of contact
(4) Outside the system

18. From a uniform square plate, one-fourth part is removed as shown. The centre of mass of remaining part will lie on



- (1) OC (2) OA (3) OB (4) OD
19. Two particles A and B initially at rest move towards each other under a mutual force of attraction. At the instant when velocity of A is v and that of B is $2v$, the velocity of centre of mass of the system
- (1) v (2) $2v$ (3) $3v$ (4) Zero
20. A ball of mass m is thrown upward and another ball of same mass is thrown downward so as to move freely under gravity. The acceleration of centre of mass is

- (1) g (2) $\frac{g}{2}$ (3) $2g$ (4) 0

21. A man of mass m starts moving on a plank of mass M with constant velocity v with respect to plank. If the plank lies on a smooth horizontal surface, then velocity of plank with respect to ground is

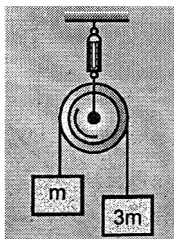
- (1) $\frac{Mv}{m+M}$ (2) $\frac{mv}{M}$
(3) $\frac{Mv}{m}$ (4) $\frac{mv}{m+M}$

SPACE FOR ROUGH WORK

22. The centre of mass of a system of three particles of masses $1g$, $2g$ and $3g$ is taken as the origin of a coordinate system. The position vector of a fourth particle of mass $4g$ such that the centre of mass of the four particle system lies at the point $(1, 2, 3)$ is $\alpha(\hat{i} + 2\hat{j} + 3\hat{k})$, where α is a constant. The value of α is :-

- (1) $\frac{10}{3}$ (2) $\frac{5}{2}$
(3) $\frac{1}{2}$ (4) $\frac{2}{5}$

23. If the system is released, then the acceleration of the centre of mass of the system :-



- (1) $\frac{g}{4}$ (2) $\frac{g}{2}$
(3) g (4) $2g$

24. A heavy nucleus at rest breaks into two fragments which fly off with velocities $8 : 1$. The ratio of radii of the fragments is :-

- (1) $1 : 2$ (2) $1 : 4$
(3) $4 : 1$ (4) $2 : 1$

25. A bullet of mass m is fired from a gun of mass M . The recoiling gun compresses a spring of force constant k by a distance d . Then the velocity of the bullet is :-

- (1) $kd\sqrt{M/m}$
(2) $\frac{d}{M}\sqrt{km}$
(3) $\frac{d}{m}\sqrt{kM}$
(4) $\frac{kM}{m}\sqrt{d}$

26. A sphere of mass m moving with a constant velocity collides with another stationary sphere of same mass. The ratio of velocities of two spheres after collision will be, if the co-efficient of restitution is e -

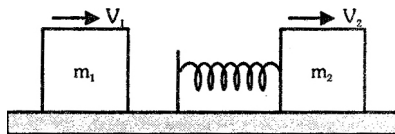
- (1) $\frac{1-e}{1+e}$ (2) $\frac{e-1}{e+1}$
(3) $\frac{1+e}{1-e}$ (4) $\frac{e+1}{e-1}$

27. A particle falls from a height ' h ' upon a fixed horizontal plane and rebounds. If V is the coefficient of restitution the total distance travelled before rebounding has stopped is :-

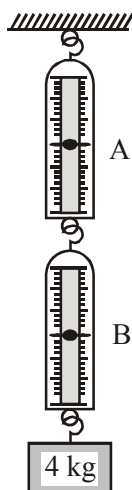
- (1) $h\left(\frac{1+e^2}{1-e^2}\right)$ (2) $h\left(\frac{1-e^2}{1+e^2}\right)$
(3) $\frac{h}{2}\left(\frac{1-e^2}{1+e^2}\right)$ (4) $\frac{h}{2}\left(\frac{1+e^2}{1-e^2}\right)$

SPACE FOR ROUGH WORK

28. Two masses $m_1 = 2\text{ kg}$ and $m_2 = 5\text{ kg}$ are moving on a frictionless surface with velocities 10 m/s and 3 m/s respectively. m_2 is ahead of m_1 . An ideal spring of spring constant $k = 1120\text{ N/m}$ is attached on the back side of m_2 . The maximum compression of the spring will be :-

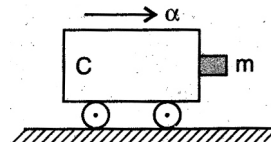


- (1) 0.51 m
(2) 0.062 m
(3) 0.25 m
(4) 0.72 m
29. A block of mass 4 kg is suspended through two light spring balances A and B. Then A and B will read respectively :-



- (1) 4 kg and zero kg
(2) zero kg and 4 kg
(3) 4 kg and 4 kg
(4) 2 kg and 2 kg

30. A block of mass m is in contact with the cart C as shown in the figure. The coefficient of static friction between the block and the cart is μ . The acceleration α of the cart that will prevent the blocks from falling satisfies:



- (1) $\alpha > \frac{mg}{\mu}$ (2) $\alpha > \frac{g}{\mu m}$
(3) $\alpha \geq \frac{g}{\mu}$ (4) $\alpha < \frac{g}{\mu}$

31. One quantum of charge should be at least be equal to the charge in coulomb.

- (1) $1.6 \times 10^{-17}\text{ C}$
(2) $1.6 \times 10^{-19}\text{ C}$
(3) $1.6 \times 10^{-10}\text{ C}$
(4) $4.8 \times 10^{-10}\text{ C}$

32. Four equal but like charge are placed at four corners of a square. The electric field intensity at the center of the square due to any one charge is E , then the resultant electric field intensity at centre of square will be.

- (1) Zero (2) $4 E$
(3) E (4) $\frac{1}{2E}$

SPACE FOR ROUGH WORK

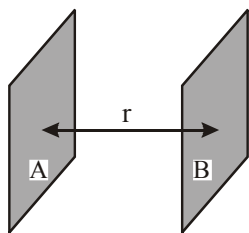
33. A charged particle of charge q and mass m is released from rest in an uniform electric field E . Neglecting the effect of gravity, the kinetic energy of the charged particle after time ' t ' seconds is.

(1) $\frac{Eqm}{t}$ (2) $\frac{E^2q^2t^2}{2m}$
(3) $\frac{2E^2t^2}{mq}$ (4) $\frac{Eq^2m}{2t^2}$

34. A small circular ring has a uniform charge distribution on a far off axial point distance x from the centre of the ring, the electric field is proportional to.

(1) x^{-1} (2) $x^{-3/2}$
(3) x^{-2} (4) $x^{5/4}$

35. The figure shows two parallel equipotential surfaces A and B kept a small distance r apart from each other. A point charge of q coulomb is taken from the surface A to B. The amount of net work done will be :



(1) $-\frac{1}{4\pi\epsilon_0} \frac{q}{r}$ (2) $\frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$
(3) $-\frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$ (4) zero

36. The electric potential V at any point (x, y, z) (all in metres) in space is given by $V = 4x^2$ volt. The electric field at the point $(1m, 0, 2m)$ in volt/metre is :

- (1) 8 along negative x-axis
(2) 8 along positive x-axis
(3) 16 along negative x-axis
(4) 16 along positive x-axis

37. An electric dipole is placed in a non-uniform electric field. Then net :

- (1) force experienced is zero while torque is not zero
(2) force experienced is zero and torque is also zero
(3) both force and torque are not zero
(4) force experience is not zero while torque is zero

38. A point charge $+q$ is placed at the mid-point of a cube of side L . The electric flux emerging from the cube is :

- (1) q/ϵ_0
(2) $q/(6L^2\epsilon_0)$
(3) $6qL^2/\epsilon_0$
(4) zero

39. A particle of mass m and charge q is placed at rest in a uniform electric field E and then released. The KE attained by the particle after moving a distance y is :

- (1) qEy^2 (2) qE^2y
(3) qEy (4) q^2Ey

SPACE FOR ROUGH WORK

40. The mathematical form of Gauss's law is :

$$\epsilon_0 \int \vec{E} \cdot d\vec{S} = q$$

In this reference which of the following is correct ?

- (1) E depends on the charge q which is enclosed within the Gaussian surface only
- (2) E depends on the charge which is inside and outside the Gaussian surface
- (3) E does not depend on the magnitude of charge q
- (4) All of the above

41. The electric field at a point at a distance r from an electric dipole is proportional to :

- (1) $\frac{1}{r}$ (2) $\frac{1}{r^2}$ (3) $\frac{1}{r^3}$ (4) r^2

42. A semi circular arc of radius a is charged uniformly and the charge per unit length is λ . The electric field at the centre is :

- (1) $\frac{\lambda}{2\pi\epsilon_0 a}$ (2) $\frac{\lambda}{2\pi\epsilon_0 a^2}$
- (3) $\frac{\lambda}{4\pi^2\epsilon_0 a}$ (4) $\frac{\lambda^2}{2\pi\epsilon_0 a}$

43. **Statement-1** : If Kinetic energy of a system is zero then linear momentum of each of particle of the system must be zero.

Statement-2 : If linear momentum of a system is zero then kinetic energy of system must be zero.

Statement-3 : If linear momentum of a particle is zero then kinetic energy of particle is zero.

Select the correct sequence.

- (1) TTF (2) TFT
- (3) FFT (4) TFF

44. A particle of mass m moving with velocity u_1 collides elastically with another particle of same mass moving with velocity u_2 in the same direction. After collision their speeds are v_1 and v_2 respectively then

- (A) $u_1 + v_1 = v_2 + u_2$
- (B) $u_1 - v_1 = v_2 + u_2$

- (1) Both the equation A and B are correct
- (2) Both the equations A and B are incorrect

- (3) Equation A is correct but not B
- (4) Equation B is correct but not A

45. Which of the following does not hold when two particles of masses m_1 and m_2 undergo elastic collision

- (1) When $m_1 = m_2$ and m_2 is stationary, there is maximum transfer to kinetic energy in head on collision.
- (2) When $m_1 = m_2$ and m_2 is stationary, there is maximum transfer to momentum in head on collision
- (3) When $m_1 \gg m_2$ and m_2 is stationary, after head on collision m_2 moves with twice the velocity of m_1
- (4) When the collision is oblique and $m_1 = m_2$ with m_2 stationary, after the collision the particle move in opposite direction.

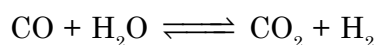
SPACE FOR ROUGH WORK

46. In a reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ degree of dissociation is 30%. If initial moles of PCl_5 is one then total moles at equilibrium is
(1) 1.3 (2) 0.7 (3) 1.6 (4) 1.0
47. For reaction $\text{HI} \rightleftharpoons \frac{1}{2} \text{H}_2 + \frac{1}{2} \text{I}_2$ value of K_c is $1/8$ then value of K_c for $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$.
(1) $\frac{1}{64}$ (2) 64 (3) $\frac{1}{8}$ (4) 8
48. In a equilibrium reaction
 $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
 $\Delta H = -3000$ calories,
which factor favours dissociation of HI :-
(1) Low temp. (2) High Pressure
(3) High temp. (4) Low pressure.
49. $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ ($\Delta H = -ve$)
If temp. of following equilibrium reaction increase then equilibrium state -
(1) Shift Right side
(2) Shift left side
(3) Unchanged
(4) Nothing say.
50. $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{CO}(\text{g})$
by increasing pressure following equilibrium
(1) Unaffected
(2) Proceed in backward direction
(3) Proceed in forward direction
(4) Unfixed
51. Unit of equilibrium constant K_c for following homogenous reaction :-
 $4\text{NH}_3 + 5\text{O}_2 \rightleftharpoons 4\text{NO} + 6\text{H}_2\text{O}$ is
(1) $(\text{Conc})^{-1}$ (2) $(\text{Conc})^{+1}$
(3) $(\text{Conc})^{+10}$ (4) Have no unit
52. Which of the following factor shifted the reaction $\text{PCl}_3 + \text{Cl}_2 \rightleftharpoons \text{PCl}_5$ at left side.
(1) Adding PCl_5
(2) Increase pressure
(3) Constant temp.
(4) Catalyst.
53. In which of the following process reaction is fastest complete.
(1) $K = 10$ (2) $K = 1$
(3) $K = 10^3$ (4) $K = 10^{-2}$
54. At 298 K equilibrium constant K_1 and K_2 of following reaction
 $\text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$ ----- (1)
 $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ ----- (2)
The relation between K_1 and K_2 is
(1) $K_1 = K_2$ (2) $K_2 = K_1^2$
(3) $K_2 = 1/K_1^2$ (4) $K_2 = 1/K_1$
55. In the following reaction
 $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ at constant temp. rate of backward reaction is increase by :
(1) Inert gas mixed at constant volume
(2) Cl_2 gas mixed at constant volume
(3) Inert gas mixed at constant pressure
(4) PCl_5 mixed in constant volume.

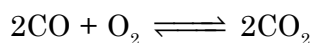
SPACE FOR ROUGH WORK

56. Some gaseous equilibrium are following :

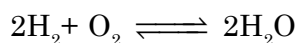
K



K_1



K_2



then find out the relation between equilibrium constants :-

- (1) $K = K_1 K_2$ (2) $K = (K_1 K_2)^2$
(3) $K = (K_1 K_2)^{-1/2}$ (4) $K = (K_1/K_2)^{1/2}$

57. For the equilibrium process $x + y \rightleftharpoons xy$. If the concⁿ. of x and y is doubled, then equilibrium constant.

- (1) Become twice (2) Become half
(3) Unchanged (4) Become thrice

58. Two moles of PCl_5 were heated in closed vessel of 2 litre capacity, at equilibrium 40% of PCl_5 was dissociated into PCl_3 and Cl_2 . The value of equilibrium constant is :

- (1) 0.267 (2) 0.53
(3) 2.67 (4) 5.3

59. The reaction $\text{A(g)} + \text{B(g)} \rightleftharpoons 2\text{C(g)}$ is occur by mixing of 3 moles of A and 1 mole of B in one litre Container. if α of B is $\frac{1}{3}$, then K_c for this reaction is :-

- (1) 0.12 (2) 0.25
(3) 0.50 (4) 0.75

60. The reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is studied in a one litre vessel at 250°C . The initial concentration of A was $3n$ and that of B was n . When equilibrium was attained, equilibrium concentration of C was found to be equal to the equilibrium concentration of B. What is the concentration of D at equilibrium :

- (1) $n/2$ (2) $(3n - 1/2)$
(3) $(n - n/3)$ (4) n

61. In a reversible reaction $\text{A} \xrightleftharpoons[k_2]{k_1} \text{B}$, the initial concentration of A and B are a and b in moles per litre and the equilibrium concentration are $(a - x)$ and $(b + x)$ respectively ; express x in terms of k_1 , k_2 , a and b :

- (1) $\frac{k_1 a - k_2 b}{k_1 + k_2}$ (2) $\frac{k_1 a - k_2 b}{k_1 - k_2}$
(3) $\frac{k_1 a - k_2 b}{k_1 k_2}$ (4) $\frac{k_1 a + k_2 b}{k_1 + k_2}$

62. The value of K_p for the reaction

$2\text{H}_2\text{O(g)} + 2\text{Cl}_2\text{O(g)} \rightleftharpoons 4\text{HCl(g)} + \text{O}_2\text{(g)}$ is 0.03 atm at 427°C , when the partial pressure are expressed in atmosphere then the value of K_c for the same reaction is :

- (1) 5.23×10^{-4} (2) 7.34×10^{-4}
(3) 3.2×10^{-3} (4) 5.43×10^{-5}

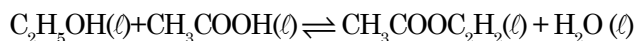
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63. The equilibrium constant of the reaction

$\text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$ is $4 \times 10^{-3} \text{ atm}^{-1/2}$. The equilibrium constant of the reaction $2 \text{SO}_3(\text{g}) \rightleftharpoons 2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ would be :

- (1) 250 atm
(2) $4 \times 10^3 \text{ atm}$
(3) $0.25 \times 10^4 \text{ atm}$
(4) $6.25 \times 10^4 \text{ atm}$

64. When alcohol ($\text{C}_2\text{H}_5\text{OH}$) and acetic acid are mixed together in equimolar ratio at 27°C , 33% is converted into ester. Then the K_c for the equilibrium :



- (1) 4 (2) 1/4 (3) 9 (4) 1/9

65. 2 moles each of SO_3 , CO , SO_2 and CO_2 is taken in a one lit. vessel.

If K_c for $\text{SO}_3 + \text{CO} \rightleftharpoons \text{SO}_2 + \text{CO}_2$ is 1/9 then :

- (1) total no. of moles at equilibrium are less than 8
(2) $n(\text{SO}_3) + n(\text{CO}_2) = 4$
(3) $[n(\text{SO}_2)/n(\text{CO})] < 1$
(4) both (2) and (3)

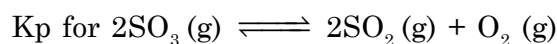
66. An equilibrium mixture in a vessel of capacity 100 litre contain 1 mol N_2 , 2 mol O_2 and 3 mol NO . No. of moles of O_2 to be added so that at new equilibrium the conc. of NO is found to be 0.04 mol/lit :

- (1) (101/18) (2) (101/9)
(3) (202/9) (4) None of these

67. Ammonia gas at 15 atm is introduced in a rigid vessel at 300 K. At equilibrium the total pressure of the vessel is found to be 40.11 atm at 300°C . The degree of dissociation of NH_3 will be :

- (1) 0.6 (2) 0.4
(3) unpredictable (4) none of these

68. The degree of dissociation of SO_3 is α at equilibrium pressure P_0 :



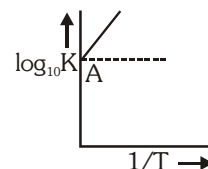
- (1) $[(P_0\alpha^3)/2(1-\alpha)^3]$
(2) $[(P_0\alpha^3)/(2+\alpha)(1-\alpha)^2]$
(3) $[(P_0\alpha^2)/2(1-\alpha)^2]$
(4) none of these

69. Given the following reaction at equilibrium

$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$. Some inert gas at constant pressure is added to the system. Predict which of the following facts will be affected :

- (1) more $\text{NH}_3(\text{g})$ is produced
(2) less $\text{NH}_3(\text{g})$ is produced
(3) no affect on the equilibrium
(4) K_p of the reaction is decreased

70. Variation of $\log_{10} K$ with $\frac{1}{T}$ is shown by the following graph in which straight line



is at 45° , hence ΔH° is :

- (1) +4.606 cal (2) -4.606 cal
(3) 2 cal (4) -2cal

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71. For an equilibrium $\text{H}_2\text{O (s)} \rightleftharpoons \text{H}_2\text{O (l)}$ which of the following statement is true :
- (1) the pressure changes do not affect the equilibrium
 - (2) more of ice melts if pressure on the system is increased
 - (3) more of liquid freezes if pressure on the system is increase
 - (4) the pressure changes may increase or decrease the degree of advancement of the reaction depending upon the temperature of the system.
72. Which of the following expressions is/are not true ?
- (1) $[\text{H}^+] = [\text{OH}^-] = \sqrt{K_w}$ for a neutral solution at all temperatures.
 - (2) $[\text{H}^+] > \sqrt{K_w}$ & $[\text{OH}^-] < \sqrt{K_w}$ for an acidic solution
 - (3) $[\text{H}^+] < \sqrt{K_w}$ & $[\text{OH}^-] > \sqrt{K_w}$ for an alkaline solution
 - (4) $[\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$ for a neutral solution at all temperatures
73. An acid solution of pH 6 is diluted thousand times. The pH of solution becomes approx-
- (1) 6.96
 - (2) 6
 - (3) 4
 - (4) 9
74. pOH of H_2O is 7.0 at 298 K. If water is heated at 350 K, which of the following should be true ?
- (1) pOH will decrease
 - (2) pOH will increase
 - (3) pOH will remain 7.0
 - (4) concentration of H^+ ions will increase but that of OH^- will decrease
75. Which of the following solution will have a pH exactly equal to 8 ?
- (1) 10^{-8} M HCl solution at 25°C
 - (2) 10^{-8} M H^+ solution at 25°
 - (3) $2 \times 10^{-6} \text{ M Ba(OH)}_2$ solution at 25° C
 - (4) 10^{-6} M NaOH solution at 50° C
76. The number of hydrogen ions in 10 mL of a solution with pH = 13 is-
- (1) 10^{13}
 - (2) 6.023×10^8
 - (3) 6.023×10^{13}
 - (4) 6.023×10^{10}
77. At 55° C autoprotolysis constant of water is 4×10^{-14} . If a given sample of water has a pH of 6.9, then it is-
- (1) acidic
 - (2) basic
 - (3) neutral
 - (4) explosive
78. A 50 mL solution of pH = 1 is mixed with a 50 mL solution of pH = 2. then pH of the mixture will be nearly-
- (1) 0.76
 - (2) 1.26
 - (3) 1.76
 - (4) 2.26
79. The pH of a solution obtained by mixing 50 mL of 0.4 N HCl and 50 mL of 0.2 N NaOH is-
- (1) $-\log 2$
 - (2) $-\log 0.2$
 - (3) 1.0
 - (4) 2.0
80. The pH of a solution is 7.00. To this solution sufficient base is added to increase the pH to 12.0. The increase OH^- ion concentration is-
- (1) 5 times
 - (2) 1000 times
 - (3) 10^5 times
 - (4) 4 times

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81. Which of the following solution will have pH close to 1.0 ?
 (1) 100 mL of M/10HCl + 100 mL of M/10 NaOH
 (2) 55 mL of M/10 HCl + 45 mL of M/10 NaOH
 (3) 10 mL of M/10 HCl + 90 mL of M/10 NaOH
 (4) 75 mL of M/5 HCl + 25 mL of M/5 NaOH.
82. Which statement/relationship is correct ?
 (1) upon hydrolysis of salt of a strong base and weak acid gives a solution with $\text{pH} < 7$
 (2) $\text{pH} = -\log \frac{1}{[\text{H}^+]}$
 (3) only at 25 °C the pH of water is 7
 (4) the value of pK_w at 25 °C is 7
83. If 50 mL of 0.2 (M) KOH is added to 40 mL of 0.5(M) HCOOH. The pH of the resulting solution is ($K = 1.8 \times 10^{-4}$) :
 (1) 3.75 (2) 5.6
 (3) 7.5 (4) 3.4
84. 50% neutralization of a solution of formic acid ($K_a = 2 \times 10^{-4}$) with NaOH would result in a solution having a hydrogen ion concentration of-
 (1) 2×10^{-4} (2) 3.7
 (3) 2.7 (4) 1.85
85. The pH of a solution obtained by mixing 100 mL of 0.2 M CH_3COOH with 100 mL of 0.2 M NaOH would be (pK_a for $\text{CH}_3\text{COOH} = 4.74$ and $\log 2 = 0.301$)
 (1) 4.74 (2) 8.87 (3) 9.10 (4) 8.57
86. Which of the following when added to 1.0 L of 0.5 M HCl would result in maximum increase in pH?
 (1) 0.5 mol CH_3COOH
 (2) 1.0 mol NaCl
 (3) 0.4 mol NaOH
 (4) 0.6 mole CH_3COONa
87. The solubility of $\text{Fe}(\text{OH})_3$ would be maximum in-
 (1) 0.1 M NaOH (2) 0.1 M HCl
 (3) 0.1 M KOH (4) 0.1 M H_2SO_4
88. Arrange in increasing order of solubility of AgBr in solutions given :
 (i) 0.1 M NH_3 (ii) 0.1 M AgNO_3
 (iii) 0.2 M NaBr (iv) pure water
 (1) (iii) < (ii) < (iv) < (i)
 (2) (iii) < (ii) < (i) < (iv)
 (3) (iii) < (ii) = (i) < (iv)
 (4) (ii) < (iii) < (iv) < (i)
89. The pH indicators are-
 (1) salts of strong acids & strong bases
 (2) salts of weak acids & weak bases
 (3) either weak acids or weak bases
 (4) either strong acids or strong bases
90. What fraction of an indicator HIn is in the basic form at a pH of 6 if pK_a of the indicator is 5 ?
 (1) $\frac{1}{2}$ (2) $\frac{1}{11}$ (3) $\frac{10}{11}$ (4) $\frac{1}{10}$

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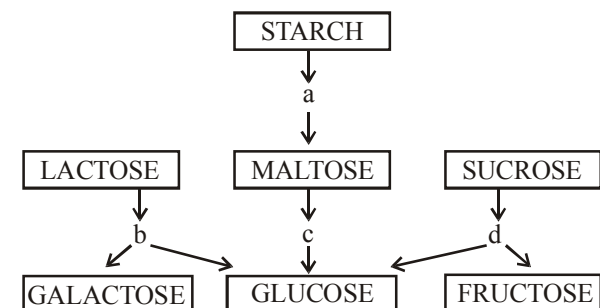
91. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
A	Goblet cells	Mucus	Prevent mucosa layer from damage of HCl
B	Lysozyme	Saliva	Antibacterial agent
C	Saliva	Subparotid gland	Secrete salivary amylase
D	Oxyntic cells	HCl	Stimulate gastric lipase

- (1) Only 'A'
(2) A and B
(3) Only C
(4) C and D
92. Which of the following statements is/are incorrect regarding digestion and absorption of food in human beings :-
- (a) About 90% of starch is hydrolysed by salivary amylase in our oral cavity
(b) Entero-endocrine cells in our stomach secrete the proenzyme trypsinogen
(c) Vitamin-D is produced in human body in skin
(d) Bile salts act as activator of pancreatic lipase
- (1) Two, a and b
(2) Two, a and c
(3) Two, a and d
(4) Three, a, b and d
93. Which of the following reaction takes place in duodenum of small intestine :-
- (1) Dipeptides $\rightarrow$ Amino acids
(2) Maltose $\rightarrow$ Glucose + Glucose
(3) Triglycerides $\rightarrow$ Monoglycerides + Fatty acid
(4) Lactose $\rightarrow$ Glucose + Glucose

94. Consider the following sequences ('a' to 'd') of histological layers of alimentary canal and select the answer from the options given below. :-
- (a) serosa, muscularis, submucosa, mucosa
(b) serosa, submucosa, muscularis, mucosa
(c) mucosa, submucosa, muscularis, serosa
(d) mucosa, muscularis, submucosa, serosa
- (1) sequence a and d
(2) sequence b and c
(3) sequence a and c
(4) sequence b and d
95. Pepsin differs from trypsin in that it digest :-
- (1) Proteins in acidic medium in stomach
(2) Proteins in alkaline medium in intestine
(3) Proteins in acidic medium in intestine
(4) Proteins in alkaline medium in stomach
96. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-
- (a) the stomach stores food for 30 minutes only
(b) Intestinal juice is also called succus entericus
(c) Mixture of food with saliva is called bolus
(d) Pancreatic juice contain digestive enzymes for digestion of proteins only.
- (1) Statements a,c,d
(2) Statements b,c
(3) Statements a,b,d
(4) Statement d only

97. Identify enzymes a, b, c and d in the digestion of carbohydrates :-



	a	b	c	d
1	Lactase	Amylase	Lactase	Invertase
2	Lipase	Lactase	Maltase	Invertase
3	Amylase	Lactase	Maltase	Sucrase
4	Amylase	Lactase	Maltase	Lipase

98. Enterokinase is :-
- (1) nondigestive enzyme secreted by duodenal serosa
 - (2) nondigestive enzyme secreted by duodenal mucosa
 - (3) nondigestive enzyme secreted by duodenal muscularis layer
 - (4) nondigestive enzyme secreted by mucosa of jejunum and ileum
99. Brunner's glands are characteristics of :-
- (1) Submucosa of duodenum
 - (2) Mucosa of jejunum
 - (3) Submucosa of stomach
 - (4) Mucosa of ileum
100. Read the following statements :-
- (I) Absorption of water depends on osmotic gradient
 - (II) Glucose and sodium ions are absorbed through active transportation
 - (III) Fructose is absorbed by simple diffusion
- Out of these which is/are not correct statements ?
- (1) Only I
 - (2) Only-II
 - (3) Only-III
 - (4) Both I & III

101. Here given the list of various digestive enzymes:-

Pepsin, Trypsinogen, ptyline, chymotrypsinogen, procarboxypeptidase, nucleosidase, nuclease, amylase, dipeptidase and Lipase.

Out of these how many enzymes are found in pancreatic juice ?

- (1) Four
- (2) Five
- (3) Six
- (4) Seven

102. Which one is not the feature of large intestine?

- (1) Some water, minerals and drugs are absorbed in it.
- (2) It stores the faeces temporarily
- (3) Digestion is not took place in it
- (4) It secretes the digestive enzymes

103. In pernicious anaemia, failure of erythrocytes to mature is mainly due to lack of :-

- (1) Vitamin-K
- (2) Thiamine
- (3) Pyridoxine
- (4) Cyanocobalamine

104. Milk protein casein $\xrightarrow{(y)}$ paracasein
 $\xrightarrow{(M)}$ calcium paracaseinate (curding of milk) :-

In the above reaction, latter (y) and (M) denote:-

- (1) Rennin and Ca^{+2} respectively
- (2) Ca^{+2} and rennin respectively
- (3) Renin, HCl and Ca^{+2} respectively
- (4) Amylase and Ca^{+2} respectively

105. Read the following statements :-

- (a) Human liver is the largest endocrine gland of body and having two lobes
- (b) Hepatic lobules are structural and functional unit of liver and contained hepatocytes which are arranged in chord like manner
- (c) Glisson's capsule is covering of each lobules and made up by connective tissue
- (d) Bile juice is formed and secreted by hepatocytes and is stored into liver sinusoid

Out of these which statements are corrects and incorrects ?

- (1) Statements a and b are correct while c & d are incorrect
- (2) Statements a and d are correct while b & c are incorrect
- (3) Statements b & c are correct while a & d are incorrect
- (4) Statements b and d are correct while a & c are incorrect

106. If for some reason the parietal cells of the stomach epithelium become partially non-functional what is likely to happen ?

- (1) The pancreatic enzymes and specially the trypsin and lipase will not work efficiently
- (2) The pH of stomach will fall abruptly
- (3) Steapsins will be more effective
- (4) Proteins will not be adequately hydrolysed by pepsin into proteoses and peptones

107. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

- (a) Serosa is outermost layer and is made up by mesothelium and connective tissue
- (b) Submucosa layer is made up of soft and areolar connective tissue, blood vessels, lymph vessels and nerve fibres.
- (c) Inner most layer of alimentary canal is made up by mucosa which forms the folds and ridges
- (d) Muscularis layer is made up by outer circular muscles and inner longitudinal muscles.

Options :-

- (1) Statements (b), (c) & (d)
- (2) Statements (a) and (d)
- (3) Statements (a), (b) & (c)
- (4) Statements (d) only

108. Consider the following four statements (a-d) and select the option which includes all the correct ones only.

- (a) On the dorsal surface of tongue, taste papillae are found and all papillae contain taste buds.
- (b) Opening of stomach into duodenum is guarded by pyloric sphincter.
- (c) Opening of oesophagus into stomach is guarded by cardiac sphincter.
- (d) Dorsal surface of tongue is connected with base of buccal cavity through frenulum linguae.

Options :-

- (1) Statements (a) and (b)
- (2) Statements (a), (b) & (c)
- (3) Statements (b) & (c)
- (4) Statements (b), (c) & (d)

109. Which one is correct dental formula of adult human ?

- (1) $I \frac{1}{1} C \frac{2}{2} Pm \frac{2}{2} M \frac{3}{3} \times 2$
- (2) $I \frac{2}{2} C \frac{1}{1} Pm \frac{2}{2} M \frac{3}{3} \times 2$
- (3) $I \frac{2}{2} C \frac{0}{0} Pm \frac{3}{3} M \frac{2}{2} \times 2$
- (4) $I \frac{2}{2} C \frac{1}{1} Pm \frac{3}{3} M \frac{2}{2} \times 2$

110. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
(i)	<i>Ptyalin enzyme</i>	<i>Salivary gland</i>	<i>Protein digesting enzyme</i>
(ii)	<i>Pepsin</i>	<i>Gastric gland</i>	<i>Protein digesting enzyme</i>
(iii)	<i>Amylase</i>	<i>Pancreas</i>	<i>Fat digesting enzyme</i>

Options :-

- (1) Only (i)
- (2) (i) & (ii) both
- (3) Only (ii)
- (4) (ii) & (iii) both

111. Given below a table in which name of digestive enzymes, their substrate and their products are given in column I, II and III respectively. Select the option in which correctly matched pairs are given ?

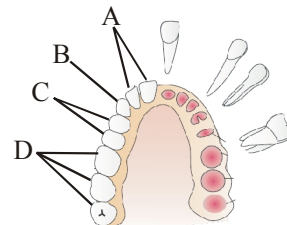
S. No.	Column-I Name of enzyme	Column-II Substrate	Column-III Product
(i)	<i>Maltase</i>	<i>Maltose</i>	<i>Glucose + Fructose</i>
(ii)	<i>Lactase</i>	<i>Lactose</i>	<i>Glucose + galactose</i>
(iii)	<i>Nucleosidase</i>	<i>Nucleotide</i>	<i>Sugar + N- Bases</i>
(iv)	<i>Sucrase</i>	<i>Sucrose</i>	<i>Glucose + Fructose</i>
(v)	<i>Nucleotidase</i>	<i>Nucleotide</i>	<i>Nucleoside + phosphate group</i>

Options :-

- (1) (i), (ii), (iv) & (v)
- (2) (i), (ii), (iii) & (v)
- (3) (ii), (iv) & (v)
- (4) (i), (iii) & (iv)

112. An arrangement of teeth on jaw bone in human beings are shown in given below diagram and different types of four teeth are labelled by alphabets A, B, C and D.

Which one of these is not correctly identified in the option, given along with its structure and function ?



		Teeth	Structure	Function
(1)	B	Canines	Long & pinpointed	Tearing & shearing
(2)	C	Premolars	Rectangular	Gnawing
(3)	D	Molars	Rectangular	Chewing
(4)	A	Incisors	Chisel shaped	Cutting / gnawing

113. Find out the correct match from the following table :-

	Column-I Name of salivary gland	Column-II Their duct	Column-III Their property
(i)	<i>Parotid gland</i>	<i>Stenson's duct</i>	<i>Largest salivary gland</i>
(ii)	<i>Sublingual gland</i>	<i>Wirsung duct</i>	<i>Smallest salivary gland</i>
(iii)	<i>Submandibular gland</i>	<i>Wharton's duct</i>	<i>Maximum salivation</i>

Options :-

- (1) Only (ii)
- (2) (i) and (iii)
- (3) Only (i)
- (4) (i), (ii) and (iii) all

114. Read the following statements :-

- (a) Chylomicrons are small lipoprotein particles that are absorbed from intestine into blood capillaries.
- (b) About 60% of starch is hydrolised by salivary amylase in our buccal cavity
- (c) Chief cells in our stomach secretes HCl and castle factor
- (d) Fructose and some amino acids are absorbed through intestinal mucosa by the help of carrier proteins

Select the option which includes only correct statement/s ?

Options :-

- (1) Statements (a) & (b)
- (2) Statements (c) & (d)
- (3) Statement (a) only
- (4) Statements (d) only

115. Which one of the following is the correct match of the site of the action on the given substrate, the enzyme acting upon it and the end product?

- (1) Duodenum $\rightarrow$ Triglycerides $\xrightarrow{\text{Trypsin}}$ Monoglycerides
- (2) Small intestine $\rightarrow$ protein $\xrightarrow{\text{Pepsin}}$ Amino acid
- (3) Stomach $\rightarrow$ Nucleic acid $\xrightarrow{\text{Nuclease}}$ nucleotide
- (4) Ileum $\rightarrow$ Dipeptides $\xrightarrow{\text{Dipeptidase}}$ Amino acid

116. The hormone which stimulate the synthesis and secretion of non enzymatic part of pancreatic juice:-

- (1) Secretin
- (2) Duocrinin
- (3) Enterocrinin
- (4) Villikin

117. Consider the following four statement (a-d) and select the option which includes all the correct ones only.

- (a) At the time of need, liver cells convert the non carbohydrates compound into glycogen is called glyconeogenesis
- (b) When glucose level in blood falls down then liver cell convert glycogen into glucose is known as glycogenolysis
- (c) Extra amount of glucose is converted into glycogen by liver cells is known as glycogenesis
- (d) Synthesis of glycogen from lactic acid by liver cell is known as glyconeogenesis

Option -

- (1) Statement (a), (b),(c) and (d)
- (2) Statement (b), (c) and (d)
- (3) Statements (a), (c) and (d)
- (4) Statements (a) and (d)

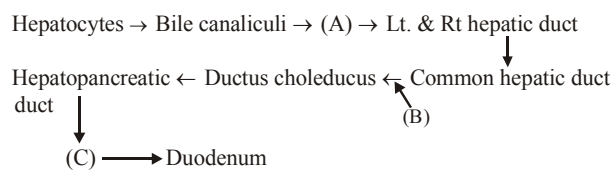
118. Read the following statements :-

- (i) "Sphinctre of oddi" controls & regulates the secretion of only pancreatic juice
- (ii) Sphinctre of boyden" controls & regulates the secretion of only bile juice
- (iii) "Sphinctre of oddi' present on ampulla of vater which controls and regulates the secretion of bile juice as well as pancreatic juice
- (iv) "Sphinctre of Boyden" present on common hepatic duct

On the basis of above statements select the correct option which includes all the correct ones ?

- (1) Only (i) & (ii)
- (2) Only (ii) & (iii)
- (3) Only (iii) & (iv)
- (4) Only (i) & (iv)

119. Given below the flow chart of pathway of bile juice in human being :-



On the basis of flow chart, select the option in which alphabets A, B and C are correctly identified according to their name ?

- (1) A = branch of hepatic duct
B = Common bile duct
C = Ampulla of vater
- (2) A = Branch of hepatic duct
B = Main pancreatic duct
C = Ampulla of vater
- (3) A = Cystic duct
B = Branch of hepatic duct
C = Main pancreatic duct
- (4) A = Branch of hepatic duct
B = Cystic duct
C = Ampulla of vater

120. Find out the correct match from the following table :-

	Column-I	Column-II	Column-III
	Name of secretion	Sources of secretion	Amount and pH
(i)	Saliva	Salivary gland	1000–1500 ml/day pH = 6.8
(ii)	Gastric juice	Gastric gland	2000 ml/day pH = 1.5 to 2.5
(iii)	Bile juice	Liver & gall bladder	500–700 ml/day pH = 7.4–7.6
(iv)	Pancreatic juice	Acini of pancreas	1000–1500 ml/day pH = 7.6 to 8.4

- (1) Only (i) & (iv)
- (2) (i) (ii) & (iv)
- (3) (i), (ii) & (iii)
- (4) All

121. Read the following statements :-

- (i) Three muscular cords are present in wall of colon are known as Teniae coli
- (ii) Haustra are the outpouching and helped in increase of absorptive surface area in ileum
- (iii) "Folds of kerkring" are present in small intestine and helped in increasing of absorptive surface area
- (iv) Ileo-caecal valve is present between colon and rectum and it regulates entry of undigested food materials into rectum.

Out of these which statements are incorrect from followings ?

- (1) (i) & (ii)
- (2) (ii) & (iii)
- (3) (i) & (iv)
- (4) (ii) & (iv)

122. "Trypsinogen" is activated by :-

- (1) HCl
- (2) Enterogastrin
- (3) Enterokinase
- (4) Pancreozymin

123. "Enamel" is originated by (A) and secreted by (B) cells. Select the option in which (A) and (B) are correctly identified ?

- (1) A = Mesoderm; B = Odontoblast
- (2) A = Ectoderm ; B = Ameloblast
- (3) A = Mesoderm; B = Osteoclast
- (4) A = Ectoderm; B = Osteoblast

124. Dipeptides are converted into amino acids by the activity of enzyme secreted by :-

- (1) Intestine
- (2) Pancreas
- (3) Stomach
- (4) Oral cavity

125. Find out the incorrect statement :-

- (1) Fats are broken down by lipase with the help of bile into di- and monosaccharides.
- (2) Mucus protects the internal mucosa from acid
- (3) Polysaccharide like starch is hydrolysed by pancreatic amylase into maltose.
- (4) Enzyme in pancreatic juice acts on nucleic acids to form nucleotides.

126. The utilisation of nutrients by the tissues for their activities is called :-

- (1) Digestion
- (2) Absorption
- (3) Assimilation
- (4) Egestion

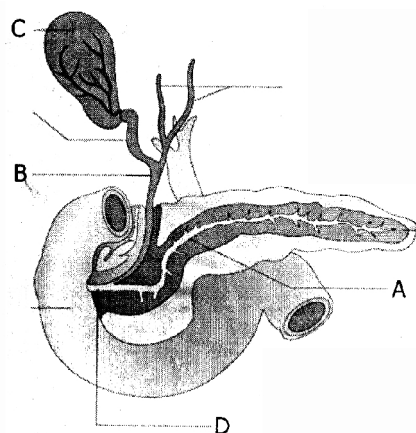
127. Which of the following layer of alimentary canal is highly vascular in nature :

- (1) Serosa
- (2) Muscularis
- (3) Submucosa
- (4) Mucosa

128. Select the duct of gall bladder :

- (1) Hepatic duct
- (2) Stenson's duct
- (3) Common hepato-pancreatic duct
- (4) Cystic duct

129. Identify A, B, C and D in the given diagram :



- (1) A = Pancreatic duct, B = Common bile duct, C = Gall bladder, D = Hepato-pancreatic duct
- (2) A = Common bile duct, B = Pancreatic duct, C = Gall bladder, D = Hepato-pancreatic duct
- (3) A = Pancreatic duct, B = Common bile duct, C = Hepato-pancreatic duct, D = Gall bladder
- (4) A = Hepato-pancreatic duct, B = Common bile duct, C = Gall bladder, D = Pancreatic duct

130. The chemical process of digestion is initiated by the action of :

- (1) Salivary Amylase
- (2) Maltase
- (3) Peptidase
- (4) Trypsin

131. The stomach stores the food for :

- (1) 1 - 2 hrs.
- (2) 2 - 3 hrs.
- (3) 3 - 4 hrs.
- (4) 4 - 5 hrs.

132. Read the statements :

- (i) The common hepatopancreatic duct is guarded by sphincter of oddi.
- (ii) The endocrine portion of pancreas secretes an alkaline pancreatic juice.
- (iii) The intrinsic factor (factor essential for absorption of Vitamin B₁₂) is secreted by oxyntic cells.
- (iv) Sub - mucosal glands are also known as Brunner's glands.

How many statements of the above are true :

- (1) 4
- (2) 3
- (3) 2
- (4) 1

133. Which one of the following is not a cell of gastric glands :

- (1) Mucus neck cells
- (2) Paneth cells
- (3) Peptic cells
- (4) Parietal cells

134. Which component of food is hydrolysed in oral cavity by the action of salivary amylase ?

- (1) fat
- (2) carbohydrate
- (3) protein
- (4) all of these

135. Which of the following is not a common symptom in both PEM-diseases Marasmus and Kwashiorkor ?

- (1) Wasting of muscle.
- (2) Protein deficiency.
- (3) Failure of growth and brain development.
- (4) Extensive oedema and swelling of body parts.

136. Match the Column I with Column II and choose the correct option :

Column - I

Column - II

- | | |
|------------------------------|---------------|
| (a) Bilirubin and Biliverdin | (i) Parotid |
| (b) Hydrolysis of starch | (ii) Bile |
| (c) Digestion of fat | (iii) Lipases |
| (d) Salivary gland | (iv) Amylases |

- (1) a = (ii), b = (iv), c = (iii), d = (i)
 (2) a = (i), b = (ii), c = (iii), d = (iv)
 (3) a = (iv), b = (iii), c = (ii), d = (i)
 (4) a = (i), b = (iv), c = (iii), d = (ii)

137. How many human teeth appear twice during the life span of an individual :

- (1) 12 (2) 20
 (3) 28 (4) 32

138. Jaundice may be caused by retarded function of :

- (1) Stomach
 (2) Liver
 (3) Intestines
 (4) Pancreas

139. Match the Column I with Column II and choose the correct option :

Column I

Column II

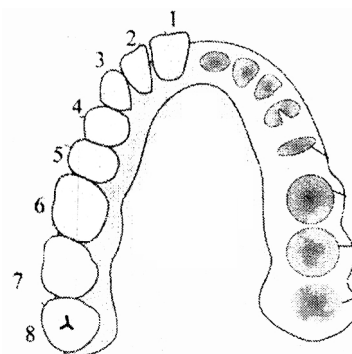
- | | |
|------------------|------------------------|
| A. Goblet cells | 1. Antibacterial agent |
| B. Lysozymes | 2. Mucus |
| C. Saliva | 3. HCl |
| D. Oxyntic cells | 4. Sublingual gland |

- (1) A = 3, B = 1, C = 4, D = 2
 (2) A = 1, B = 3, C = 4, D = 2
 (3) A = 2, B = 3, C = 1, D = 4
 (4) A = 2, B = 1, C = 4, D = 3

140. Regarding the digestive system which is true :

- (1) The bases of villi in the intestine have crypts of langerhans.
 (2) The cystic duct directly opens into duodenum and carries bile.
 (3) The bile helps in emulsification and also helps in activity of lipase.
 (4) The exocrine part of pancreas secretes the acidic pancreatic juice.

141. The diagram of human jaw is given showing the teeth. The names of teeth numbered 3 and 4 are :-



- (1) Incisor and Canine
 (2) Canine and Premolar
 (3) Premolar and molar
 (4) Incisor and Premolar

142. Which of the following is not a disorder of digestive system :

- (1) Jaundice
 (2) Vomiting
 (3) Angina pectoris
 (4) Constipation

143. Match the following :

Column I

Column II (pH)

- | | |
|-----------------------|---------|
| I. Acid in stomach | (a) 6.8 |
| II. Salivary amylase | (b) 7.8 |
| III. Pancreatic juice | (c) 1.8 |

- (1) I = a, II = b, III = c
 (2) I = b, II = a, III = c
 (3) I = c, II = a, III = b
 (4) I = c, II = b, III = a

144. Bile is formed in :

- (1) Gall bladder
 (2) Liver
 (3) Spleen
 (4) Blood

145. Proteolytic enzymes do not corrode lining of alimentary canal as :

- (1) They are secreted in inactive form
 (2) Lining layer of alimentary canal does not contain protein
 (3) The enzymes are not capable of digesting fat
 (4) None of these

146. In stomach which one comes :

- (1) Bolus (2) Chyme
(3) Chyle (4) All

147. Which one correctly explains the condition of constipation :

- (1) The food is not properly digested leading to a feeling of fullness.
(2) The faeces are retained within the rectum as the bowel movements occur irregularly.
(3) It is the ejection of stomach contents through the mouth.
(4) The abnormal frequency of bowel movement and increased liquidity of the faecal discharge.

148. Where do certain symbiotic microorganism normally occurs in human body ?

- (1) Vermiform appendix and rectum
(2) Duodenum
(3) Caecum
(4) Oral lining and tongue surface

149. Emulsification of fat will not occur in the absence of :

- (1) Lipase
(2) Bile pigments
(3) Bile salts
(4) Pancreatic juice

150. The distal part of the stomach that opens into duodenum is called :

- (1) Fundus (2) Pylorus
(3) Omentum (4) Cardiac

151. Which of the following is the wrong match in the given table?

Options :

		Parotid gland	Submaxillary gland	Sublingual gland
1	Location	Below ear	Jaw angle	Below tongue
2	Number	One pair	One pair	One pair
3	Duct	Stenson duct	Duct of Rivinus	Wharton duct
4	Secretion	Saliva + enzyme	Saliva	Saliva

152. The structural and functional unit of liver is :

- (1) Glisson's capsule
(2) Hepatic lobules
(3) Hepatic vein
(4) Hepatic cords

153. Which part of alimentary canal does not secrete any digestive enzyme ?

- (1) Mouth (2) Stomach
(3) Duodenum (4) Oesophagus

154. The epithelial cells lining the stomach of vertebrates is protected from the damage by HCl because :

- (1) HCl is too dilute
(2) HCl is neutralized by stomach
(3) The epithelial cells are resistant to the action of HCl
(4) The epithelial cells are covered by a mucous secretion.

155. Which of the following correctly shows the composition of saliva ?

- (1) Electrolytes, Salivary amylase and Lipase
(2) Electrolytes, Dipeptidase and Lysozyme
(3) Electrolytes, Salivary amylase and Lysozyme
(4) Salivary amylase, Lipase and Lysozyme

156. Pulp cavity of teeth is surrounded by :

- (1) Enamel (2) Gums
(3) Bones (4) Odontoblasts

157. Which one of these are most essential for the body growth and formation of new cells ?

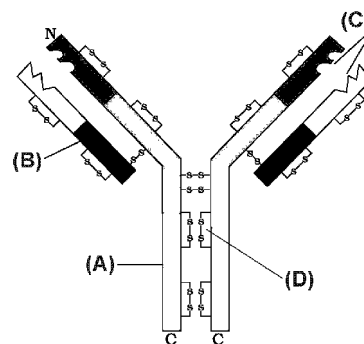
- (1) Fats (2) Sugars
(3) Vitamins (4) Proteins

158. Herbivorous animals can digest cellulose because-

- (1) they have large alimentary canal
(2) Bacteria present in their caecum help in the digestion of cellulose.
(3) Gastric juice has a digestive enzyme for cellulose digestion
(4) their molar and premolar teeth can crush and grind the food

- 159** Peyer's patches are found in
- (1) Lamina propria of ileum
 - (2) submucosa of duodenum
 - (3) Serosa of jejunum
 - (4) Muscularis of stomach
- 160.** The stomach does not digest itself because
- (1) Gastric mucosa is protected by thick layer of mucus
 - (2) Gastric mucosal cells are not digestible
 - (3) Gastric mucosal cells transport hydrogen ions out of the gastric mucosa
 - (4) Hydrogen ions are completely neutralized by mucus
- 161.** Hepatitis 'B' vaccine is produced by recombinant DNA technology. In this technology antigenic polypeptide of pathogen is produced in bacteria or yeast. So this vaccine is an example of -
- (1) I generation vaccine
 - (2) II generation vaccine
 - (3) III generation vaccine
 - (4) None of these
- 162.** Dr. Jenner made two shallow cuts on James phipp's arm and inoculated them with matter taken from cowpox pustules. Did this make James immune to small pox ?
- (1) No, because small pox and cow pox are antigenecally same.
 - (2) No, because small pox and cow pox are antigenecally different
 - (3) Yes, because small pox and cow pox are caused by such type of bacteria that are very simillar in antigenic properties.
 - (4) None of these
- 163.** Which property of immune system is basis of vaccination ?
- (1) Specificity
 - (2) Diversity
 - (3) Differentiation
 - (4) Memory

- 164.** The overall ability of the to fight the by a system is called :
- (1) host, pathogen, immunity
 - (2) pathogen, host, immunisation
 - (3) Immunity, host, pathogen
 - (4) host, immunity, vaccination
- 165.** Innate immunity is :
- (1) Specific
 - (2) Non-specific
 - (3) Sometimes specific
 - (4) (2) & (3) both
- 166.** Mucosal associated lymphoid tissue(MALT) is found in all except :
- (1) Digestive syetem
 - (2) Respiratorysystem
 - (3) Urogenital tract
 - (4) Heart
- 167.** PMNL and natural killer cells are example of which type of barrier :
- (1) Physiological
 - (2) Cellular
 - (3) Physical
 - (4) None
- 168.** Identify the structure A, B, C & D in antibody molecule:



- (1) A = Heavy Chain, B = Light chain, C = Antigen binding site, D = disulphide bond
- (2) A = Heavy Chain, B = Light chain, C = disulphide bond, D = Antigen binding site
- (3) A = Light chain, B = Heavy Chain, C = disulphide bond, D = Antigen binding site
- (4) A = Antigen binding site, B = Heavy Chain, C = disulphide bond, D = Light chain

169. Which immune response is quick and highly intensified:
(1) Primary (2) Secondary
(3) Anamnestic (4) (2) & (3) both
170. Small proteins produced by vertebrate cells naturally in response to viral infections and which inhibit the multiplication of viruses are called :
(1) Immunoglobulins
(2) Interferons
(3) Antitoxins
(4) Lipoproteins
171. Characters of acquired immunity are :
(1) Specificity
(2) Discriminate between self and non self
(3) Retains memory
(4) All of the above
172. A localised inflammatory response appears at the site of infection causes redness, swelling, pain and heat due to certain chemicals, they are :
(1) Histamine and prostaglandins
(2) Cerumen and mucus
(3) Histamine and cerumen
(4) prostaglandins and cerumen
173. The function of IgE is :
(1) Mediate in allergic response
(2) Activation of B-cells
(3) Protection from inhaled/ingested pathogens
(4) Stimulation of complement system, passive immunity to foetus
174. A person likely to develop tetanus is immunised by administering :
(1) Dead germs
(2) Preformed antibodies
(3) Wide spectrum antibodies
(4) Weakened germs
175. How does vaccination work :
(1) The immune system produces antibodies which not stay in blood
(2) Memory lymphocyte are produced. They remain in the blood to fight off any future infection with the live pathogen
(3) The dead pathogen stay in the body and constantly stimulate immune system
(4) All of these

176. In which one of the following options, the two examples are correctly matched with their particular type of immunity :

Example	Type of immunity
(1) Polymorphonuclear leucocytes and monocytes	Cellular barrier
(2) Anti-tetanus and anti-snake bite injection	Active immunity
(3) Saliva in mouth and tears in eyes	Physical barriers
(4) Mucus coating of epithelium of urino-genital tract and HCl in stomach	Physiological barrier

177. Short lived immunity acquired from mother to foetus across placenta or through mother's milk to the infant is characterized as :
(1) Active immunity
(2) Passive immunity
(3) Cellular immunity
(4) Innate non-specific immunity

178. Which of the following organs is primarily concerned with immunity :
(1) Liver
(2) Lymphatic tissue
(3) Kidney
(4) Thyroid

179. The cells which suppress the entire immune system from its attack in the same body are known as :
(1) Helper T-cells
(2) killer B-cells
(3) Memory cells
(4) Suppressor T-cells

180. Interferons are -
(1) Anti bacterial proteins
(2) Anti viral proteins
(3) Anti helminthic proteins
(4) Anti fungal proteins

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1



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 AIR-69 PRIYANSHU AGARWAL Classroom	 AIR-70 ISHIKA KAMRA Classroom	 AIR-71 STUTI KHANDWALA Classroom	 AIR-72 NAVEN SHARMA Distance	 AIR-75 SARTHAK SHARMA Classroom	 AIR-76 VANSHIKA ARORA Distance	 AIR-79 SAIYAM V. BHATT Classroom	 AIR-81 UDITH MOHAN Distance	 AIR-84 PRAJNAPAN BASU Distance	 AIR-85 HARSH RAMOLIYA Distance	 AIR-86 UTKARSH SHARMA Classroom	 AIR-87 NISHTHA Distance	 AIR-89 ARYAN BANSAL Distance	 AIR-91 SANGEETA AGRAWAL Classroom	 AIR-92 YASHDEEP Classroom	 AIR-93 ISHAN GUPTA Classroom	
 AIR-94 MEGHA GUPTA Classroom	 AIR-96 GAUTAM KAUSHAL Classroom	 AIR-97 POOJA YADAV Distance													 AIR-99 PRAGYA MITRA Distance	 AIR-100 VIVEK KUMAR Classroom

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